# Week 1 - Domain 1: SDLC Automation - AWS CodeConnections (GitHub / Bitbucket / GitLab)

**Study week:** Oct 19 - Oct 25, 2026 **Exam:** AWS DOP-C02 (Domain 1 = 22% of the exam)

| Section | What it covers |
|---|---|
| 6.1 - 6.3 | What CodeConnections is, supported providers, how a connection works (states, hosts, scope) |
| 6.4 - 6.5 | Services that use connections, the CodePipeline source action |
| 6.6 - 6.9 | IAM permissions, CodeBuild, CloudFormation Git sync, infrastructure as code |
| 6.10 - 6.11 | Troubleshooting, security best practices |
| 6.12 - 6.14 | Exam traps, hands-on lab, check yourself |

## Part 6: AWS CodeConnections

### 6.1 What CodeConnections is

- **A managed, secure link between AWS and a Git provider outside AWS.** CodePipeline, CodeBuild and other AWS services can **read your repositories and react to pushes and pull requests** without you storing a personal access token or password in AWS.
- **Renamed in 2024:** formerly **AWS CodeStar Connections**, now **AWS CodeConnections**. The exam and older docs may use either name.

| | Old name | New name |
|---|---|---|
| Service | CodeStar Connections | CodeConnections |
| Connection ARN | arn:aws:codestar-connections:... | arn:aws:codeconnections:... |
| IAM action prefix | codestar-connections: | codeconnections: |

- Both prefixes work; new connections get the new ARN format. IAM policies may need **both prefixes** (`codestar-connections:UseConnection` **and** `codeconnections:UseConnection`).
- **No extra charge** for connections.
- **Replaces:** the **GitHub version 1** source action (OAuth / personal access token) - **deprecated** - and self-managed webhooks (`PutWebhook` API) - legacy.

### 6.2 Supported providers

| Provider | Type | What it needs |
|---|---|---|
| GitHub (github.com, incl. GitHub Enterprise Cloud) | Cloud | A connection only |
| GitHub Enterprise Server | Self-managed | A host resource + the connection |
| Bitbucket Cloud | Cloud | A connection only |
| GitLab.com | Cloud | A connection only |
| GitLab self-managed | Self-managed | A host resource + the connection |

- **Not supported:** e.g. **Azure DevOps Repos**, **Bitbucket Server / Data Center**. Workaround: the provider's own CI zips the code and uploads it to **S3** (S3 source action), or mirrors to **CodeCommit**.

### 6.3 How a connection works

1. **Create** the connection (console: Developer Tools > Settings > Connections, CLI, or CloudFormation).
2. **Complete the handshake** with the provider:

  - **GitHub:** installs the **"AWS Connector for GitHub" GitHub App** in your account or organization - choose **all repositories or only selected ones**.
  - **Bitbucket:** the AWS app is installed in your **workspace**.
  - **GitLab:** you authorize access to your GitLab account or group.
3. The connection becomes **AVAILABLE**. The provider sends push and pull request events through the app's webhooks - you don't create webhooks yourself.

**Connection states:**

| State | Meaning |
|---|---|
| PENDING | Created but not yet authorized - pipelines using it fail |
| AVAILABLE | Ready to use |
| ERROR | Something broke - e.g. app uninstalled or access revoked |

- **Most-tested point:** a connection created with **CloudFormation or the CLI always starts PENDING**. Someone must open it in the **console**, choose **"Update pending connection"** and finish the sign-in. **This can't be fully automated** (the provider requires a person to approve).
- **GitHub organizations:** an **org owner** may need to approve the app. **Bitbucket:** a **workspace admin** may need to approve.

**Hosts (GitHub Enterprise Server, GitLab self-managed):**

- A **host** holds your server's **URL** and optional network settings.
- **Private network servers:** host **VPC configuration** = VPC ID, **subnets**, **security groups**, optional **TLS certificate** (self-signed / private CA). AWS reaches the server through network interfaces in your VPC.
- A host also starts **PENDING** - finish setup in the console, then create the connection using the host.

**Scope and ownership:**

- A connection belongs to **one account and one Region**. The pipeline / CodeBuild project must use a connection **in its own account and Region**.
- One app installation on the provider can be used by several connections (e.g. one per account).
- **Revoke access:** delete the connection in AWS, or uninstall the app / remove repositories on the provider side.

## 6.4 Services that use connections

| Service | How it uses the connection |
|---|---|
| CodePipeline | Source action (provider CodeStarSourceConnection) for GitHub, GitHub Enterprise Server, Bitbucket, GitLab |
| CodeBuild | Source credentials for GitHub / Bitbucket / GitLab projects and webhooks; full clone in a pipeline (CODEBUILD_CLONE_REF) |
| CloudFormation Git sync | Keeps a stack in sync with a template file in Git |
| Service Catalog | Syncs products from a Git repository |
| CodeGuru Reviewer | Reviews pull requests on GitHub and Bitbucket |
| Others | AWS Proton, Amazon Q Developer, other developer tools |

## 6.5 CodePipeline source action

```
{
  "name": "Source",
  "actionTypeId": { "category": "Source", "owner": "AWS", "provider": "CodeStarSourceConnection", "version":
     "1" },
  "configuration": {
    "ConnectionArn": "arn:aws:codeconnections:us-east-1:111122223333:connection/abc123",
    "FullRepositoryId": "my-org/payments-service",
    "BranchName": "main",
    "OutputArtifactFormat": "CODE_ZIP",
    "DetectChanges": "true"
  },
  "outputArtifacts": [{ "name": "SourceOutput" }],
  "namespace": "SourceVariables"
}
```

| Setting | Meaning |
|---|---|
| ConnectionArn | Which connection to use |
| FullRepositoryId | owner/repository, e.g. my-org/payments-service |
| BranchName | The branch to build |
| OutputArtifactFormat | CODE_ZIP (default, files only, no .git) or CODEBUILD_CLONE_REF (full git clone with history and tags) |
| DetectChanges | true (default) = start automatically on push; false = start only manually / by API |

- **Output variables:** `CommitId`, `CommitMessage`, `BranchName`, `FullRepositoryName`, `ConnectionArn`, `AuthorDate` (e.g. `#{SourceVariables.CommitId}`).
- **V2 triggers (connection sources only): push** triggers with **branch**, **file path** and **Git tag** filters (include / exclude); **pull request** triggers on **opened, updated, closed** with branch and file path filters.
- Monorepo folder = file path filter. Release on `v*` tag = Git tag trigger. Validate PRs into main = pull request trigger.

## 6.6 IAM permissions

| Who | Needs |
|---|---|
| Pipeline service role | UseConnection (codestar-connections: and codeconnections:) on the connection ARN |
| CodeBuild service role (CODEBUILD_CLONE_REF or connection source) | UseConnection too - otherwise the clone fails with access denied |
| Person / tool creating the pipeline | PassConnection, plus GetConnection / ListConnections |
| Admins managing connections | CreateConnection, DeleteConnection, UpdateConnectionInstallation, CreateHost... |

**Limit a role to certain repositories:**

```
{
  "Effect": "Allow",
  "Action": ["codestar-connections:UseConnection", "codeconnections:UseConnection"],
  "Resource": "arn:aws:codeconnections:us-east-1:111122223333:connection/abc123",
  "Condition": {
    "ForAllValues:StringEquals": {
      "codestar-connections:FullRepositoryId": ["my-org/payments-service"]
    }
  }
}
```

- Useful condition keys: `FullRepositoryId` (specific repositories), `ProviderPermissionsRequired` (`read_only` / `read_write`), `ProviderAction`, `ProviderType` (GitHub, Bitbucket, GitLab), `PassedToService` (e.g. only `codepipeline.amazonaws.com`).

## 6.7 CodeBuild with connections

- **Source credentials:** GitHub / Bitbucket / GitLab projects can use OAuth, a personal access token, or a **CodeConnections connection** (recommended - no token stored). A connection can be the **default credential** for the account and Region.
- **Webhooks:** CodeBuild projects with these sources can start builds from **webhooks with filter groups** (event type, branch, file path, actor) **without CodePipeline** - e.g. pull request checks with build status reported back to the provider.
- **CodeBuild-hosted GitHub Actions runners:** CodeBuild runs **GitHub Actions workflow jobs** on AWS-managed compute, through a webhook on the `WORKFLOW_JOB_QUEUED` event - useful when jobs need private VPC resources, IAM roles, or bigger machines.

## 6.8 CloudFormation Git sync

- **Keeps a CloudFormation stack in sync with a Git repository.** A push that changes the **template** or **deployment file** on the chosen branch updates the stack automatically - no pipeline.
- The **deployment file** (YAML) lists the template path, parameters and tags.
- Needs a **connection**, a **repository link** (`AWS::CodeStarConnections::RepositoryLink`) and a **sync configuration**.
- Good for simple GitOps; use a full pipeline when you need tests, approvals or several stages.

## 6.9 Infrastructure as code for connections

- CloudFormation: `AWS::CodeStarConnections::Connection` with `ConnectionName` and `ProviderType` (GitHub, Bitbucket, GitLab...) **or** `HostArn`. **Still created PENDING** - authorize once in the console.
- **Pattern:** create the connection once per account, authorize it, store its ARN in **Parameter Store** (or a stack output), and reference that ARN from all pipeline stacks.

## 6.10 Troubleshooting

| Symptom | Cause and fix |
|---|---|
| Source stage fails; connection PENDING | Complete the handshake in the console (Update pending connection) |
| Connection status ERROR | App uninstalled, access revoked, or provider token invalid - update installation or recreate |
| Repository missing from the list | App installed for selected repositories only - add the repository to the app installation |
| Can't install the app in the organization | Org owner (GitHub) or workspace admin (Bitbucket) must approve |
| Pipeline: not authorized to use the connection | Add UseConnection (both prefixes) to the pipeline role |
| CodeBuild clone fails with CODEBUILD_CLONE_REF | Add UseConnection to the CodeBuild role |
| Creating a pipeline fails: access denied on connection | The creator needs PassConnection |
| Pipeline doesn't start on push | DetectChanges false, V2 trigger filter doesn't match, or connection not AVAILABLE |
| Private GitHub Enterprise Server unreachable | Host with VPC configuration (subnets, security groups, TLS certificate) |
| Policy with only codestar-connections: fails on a new connection | Add the codeconnections: actions and ARN format too |

## 6.11 Security best practices

- **No long-lived tokens** stored in AWS; access comes through the provider's app and can be revoked any time.
- Give the app access only to the **repositories it needs** (selected repositories).
- Limit roles with `FullRepositoryId` and `PassedToService` conditions.

- Only **admins** create, update or delete connections; developers get `UseConnection` / `PassConnection` only where needed.
- **CloudTrail** records connection API calls.
- Self-managed servers on a private network: **host with VPC configuration** so traffic doesn't cross the internet.

## 6.12 Exam traps: quick table

| Scenario | Answer |
|---|---|
| GitHub source without storing tokens | CodeConnections source action (not GitHub v1 OAuth / PAT) |
| CloudFormation-created pipeline fails; connection PENDING | Complete the handshake in the console (can't be automated) |
| GitHub Enterprise Server in a private network | Host with VPC configuration |
| Repository in Azure DevOps or Bitbucket Server | Not supported - push a zip to S3 (or mirror to CodeCommit) |
| Build needs git log or tags | CODEBUILD_CLONE_REF + UseConnection on the CodeBuild role |
| Role should only use one repository | codestar-connections:FullRepositoryId condition |
| User can't create a pipeline with the connection | Needs PassConnection |
| Pipeline role denied on a new connection ARN | Add codeconnections:* actions (new prefix) |
| Monorepo folder filter, tag releases, PR triggers | V2 triggers (connection sources only) |
| PR builds with status on GitHub, no pipeline | CodeBuild webhook with filter groups |
| GitHub Actions jobs on AWS compute with VPC access | CodeBuild-hosted GitHub Actions runners |
| Keep a stack automatically in sync with Git | CloudFormation Git sync |
| Repository not visible when creating the source action | Add it to the app installation (selected repositories) |
| Pipeline in account B needs account A's connection | Not possible - create a connection in account B |

## 6.13 Hands-on lab (about 1 hour)

1. Console: Developer Tools > Settings > Connections - create a **GitHub** connection; install the app on **selected repositories**.
2. Create a second connection with the CLI: `aws codeconnections create-connection --provider-type GitHub --connection-name cli-test`. Confirm it is **PENDING**, then finish it with **Update pending connection**.
3. Create a **V2 pipeline** with the connection source, namespace `SourceVariables`, and `CODEBUILD_CLONE_REF`. Run `git log -1` in CodeBuild. Remove `UseConnection` from the CodeBuild role, watch it fail, then add it back.
4. Add a **push trigger** with a file path filter and a **pull request trigger**; open a PR on GitHub and watch the pipeline start.
5. Restrict the pipeline role with a **FullRepositoryId** condition; point the pipeline at another repository and confirm it is denied.
6. Optional: **CloudFormation Git sync** for a small stack (one S3 bucket); change the template in Git.
7. Clean up: pipeline, connections, stack; uninstall the GitHub app (GitHub > Settings > Applications).

## 6.14 Check yourself

1. A connection created by CloudFormation stays PENDING and the pipeline fails. How do you fix it?
2. GitHub Enterprise Server runs in a private VPC. What do you need?
3. A CodeBuild action fails to clone with CODEBUILD_CLONE_REF, but the pipeline source succeeded. Which permission is missing, on which role?
4. How do you allow a pipeline role to use only `my-org/payments` through a shared connection?
5. Your repositories are in Azure DevOps. Can CodeConnections be the pipeline source?

**Answers:**

1. Open the connection in the console, choose **Update pending connection** and finish signing in with the provider.
2. A **host** with a **VPC configuration** (VPC, subnets, security groups, TLS certificate if needed), and a connection that uses the host.
3. **UseConnection** (`codestar-connections:` and `codeconnections:`) on the **CodeBuild service role**.
4. **UseConnection** with the condition `codestar-connections:FullRepositoryId` **=** `my-org/payments`.
5. **No.** Have Azure DevOps upload a zip to **S3** (S3 source action), or mirror to CodeCommit.